

# Nhat-Minh Nguyen, Ph.D.

Cosmologist, Kavli IPMU Fellow

Kavli Institute for Physics and Mathematics of the Universe, University of Tokyo

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## RESEARCH PROFILE

Cosmologist working on growth of large-scale structure, new physics from galaxy surveys, and statistical methods for extracting information from cosmological data. Current research connects field-level inference, effective-field-theory likelihoods, multi-tracer analyses, and machine-learning methods for spectroscopic surveys. Publication record: 21 papers, including 15 outside large-collaboration author lists; h-index 17 on the current CV.

## SELECTED HONORS

**Buchalter Cosmology Prize**, with American Astronomical Society award announcement 2024

## GRANTS

**JSPS KAKENHI Grant-in-Aid for Research Activity Start-up, PI, JPY 2,000,000** 2025

**Japan Foundation for Promotion of Astronomy Research Support Grant, PI, approx. JPY 320,000** 2025

## SCIENTIFIC LEADERSHIP

**Beyond-2-Point Statistics Meet Survey Systematics** 2025  
Organizing Committee Chair

**Cosmology 2025** 2025  
Scientific Program Committee member

**COSMO'24** 2024  
Large-scale Structure Session Convener

**Future of Artificial Intelligence for Science in Japan** 2024  
Cosmology/Astrophysics Unconference Facilitator

**Michigan Cosmology Summer School** 2023  
Local Organizing Committee member

## MENTORING AND TEACHING

- Co-mentored Carter Matties at the University of Michigan; now Physics Graduate Student at Syracuse.
- Co-mentored Andrija Kostić at the Max Planck Institute for Astrophysics; now Research Scientist at DeepL.
- Advised research projects for Baptiste Barthe-Gold, Eleni Tsaprazi, Andrew Hope, Kyle Lee, Disha Saxena, and Kasey Thai.
- Course instructor, Michigan Math and Science Scholars, “Climbing the Distance Ladder to the Big Bang: How astronomers survey the Universe,” Summer 2024.
- Guest lecturer, Michigan Math and Science Scholars, Summer 2023.

## POSITIONS

**Kavli IPMU Fellow** Oct. 2024 – present  
*Kavli Institute for Physics and Mathematics of the Universe, University of Tokyo*

**Leinweber Fellow** Jan. 2022 – Jul. 2024  
*Leinweber Center for Theoretical Physics and Physics Department, University of Michigan*

**Research Associate** Jul. 2020 – Dec. 2021  
*Physical Cosmology Group, Max Planck Institute for Astrophysics*

## EDUCATION

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**Ph.D., Astrophysics, Magna Cum Laude**

2016 – 2020

International Max Planck Research School on Astrophysics; Ludwig Maximilians University of Munich; Max Planck Institute for Astrophysics

**M.Sc., Astronomy and Astrophysics**

2014 – 2016

Erasmus Joint Master Degree in Astronomy & Astrophysics, with full-tuition scholarship and stipend

**B.Sc., Physics and Theoretical Physics**

2008 – 2013

Ho Chi Minh University of Science, with honors and scholarships

## SURVEY COLLABORATIONS

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**Current**

Prime Focus Spectrograph (PFS)

**Former**

Dark Energy Spectroscopic Instrument (DESI)

## RESEARCH THEMES RELEVANT TO JPL

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- Field-level and forward modeling for galaxy clustering, intrinsic alignment, and initial-condition inference.
- Effective-field-theory likelihoods for large-scale structure and consistency tests of physical data models.
- Multi-tracer and multi-survey calibration with PFS–DESI overlap.
- Growth-of-structure tests, modified-gravity parameterizations, and dark-sector phenomenology.
- Beyond-two-point galaxy clustering statistics, mock-data challenges, and survey-analysis validation.
- Machine-learning and statistical reconstruction of the local density field.

## SELECTED PUBLICATIONS

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1. **Nguyen**. “Multi-tracers, multi-surveys: data-driven EFT prior calibration from the PFS–DESI overlap.” [arXiv preprint](#). Submitted to JCAP, 2026.
2. **Nguyen**, Akitsu, Taruya. “Galaxy sizes as complementary (zero-)bias tracers of local primordial non-Gaussianity.” [arXiv preprint](#). Submitted to PRD, 2026.
3. Barthe-Gold, **Nguyen**, Thiele. “Reconstructing the local density field with combined convolutional and point cloud architecture.” [arXiv preprint](#). Accepted for NeurIPS 2025 Workshop ML4PS.
4. Beyond-2point Collaboration. “A parameter-masked mock data challenge for beyond-2pt galaxy clustering statistics.” *ApJ* 990, 99, 2025.
5. **Nguyen**, Schmidt, Tucci, Reinecke, Kostić. “How much information can be extracted from galaxy clustering at the field level?” *Phys. Rev. Lett.* 133, 221006, 2024.
6. Wen, **Nguyen**, Huterer. “Sweeping through Horndeski Canvas: a New Growth-Rate Parameterization for Modified-Gravity Theories.” *JCAP* 09, 028, 2023.
7. **Nguyen**, Huterer, Wen. “Evidence for suppression of structure growth in the concordance cosmological model.” *Phys. Rev. Lett.* 131, 111001, 2023. Editor’s Suggestion.
8. Kostić, **Nguyen**, Schmidt, Reinecke. “Consistency tests of field level inference with the EFT likelihood.” *JCAP* 07, 063, 2023.
9. Tsaprazi, **Nguyen**, et al. “Field-level inference of galaxy intrinsic alignment from the SDSS-III BOSS survey.” *JCAP* 08, 003, 2022.
10. **Nguyen** et al. “Impacts of the physical data model on the forward inference of initial conditions from biased tracers.” *JCAP* 03, 058, 2021.
11. **Nguyen** et al. “Taking measurements of the kinematic Sunyaev-Zel’dovich effect forward: including uncertainties from velocity reconstruction with forward modeling.” *JCAP* 12, 011, 2020.
12. DESI Collaboration. “DESI 2024 VII: Cosmological Constraints from the Full-Shape Modeling of Clustering

Measurements.” JCAP 07, 028, 2025.

13. DESI Collaboration. “DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations.” JCAP 02, 021, 2025.
14. Schmidt et al. “A rigorous EFT-based forward model for large-scale structure.” JCAP 01, 042, 2019.

## SELECTED TALKS

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*Conference and workshop talks only; ordinary seminar talks are intentionally omitted.*

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|-------------|------------------------------------------------------------------------------------------------------------------|
| <b>2025</b> | Particles, Gravitation and the Universe: from Quantum Mechanics to Quantum Gravity, IOP, VAST, Ha Noi, Viet Nam. |
| <b>2024</b> | “Theory and Data Analysis Challenges for Cosmological Large-Scale Structure Observations,” YITP, Kyoto, Japan.   |
| <b>2024</b> | COSMO’24, Kyoto, Japan.                                                                                          |
| <b>2024</b> | PASCOS 2024, Plenary Session, ICISE Quy Nhon, Viet Nam.                                                          |
| <b>2024</b> | Cosmology from Home, online conference.                                                                          |
| <b>2024</b> | DESI Research Forum.                                                                                             |
| <b>2024</b> | Aspen workshop, “Fundamental Physics in the Era of Big Data and Machine Learning.”                               |
| <b>2024</b> | “Fundamental Physics from Future Spectroscopic Surveys,” LBNL, Berkeley.                                         |
| <b>2024</b> | CCA Cosmology Meeting, Flatiron Institute.                                                                       |
| <b>2023</b> | Cosmology from Home, online conference.                                                                          |
| <b>2023</b> | Future Science with CMB x LSS workshop, YITP, Kyoto.                                                             |
| <b>2022</b> | Aspen workshop, “Large-Scale Structure Cosmology beyond 2-Point Statistics.”                                     |

## POPULAR SCIENCE AND MEDIA

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| <b>2024</b>  | Vietnamese-language article on DESI and dark energy, Tia Sang Magazine.                                                                                                       |
| <b>2023</b>  | “Growing in molasses: Cosmic large-scale structure caught growing slower than expected,” Science X Dialog, Phys.org.                                                          |
| <b>2023</b>  | Vietnamese-language article on searching for new signals from the Universe, Tia Sang Magazine.                                                                                |
| <b>Media</b> | Coverage in Scientific American, New Scientist, VICE, Space.com, Quanta Magazine, University of Michigan News, and Max Planck Institute for Astrophysics Research Highlights. |

## PROFESSIONAL SERVICE

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| <b>Referee</b>   | Physical Review Letters, Astrophysical Journal Letters, Physical Review D, Journal of Cosmology and Astroparticle Physics, Astronomy & Astrophysics.                                                                                       |
| <b>Community</b> | Founder or main organizer of VLLT Joint Astronomy & Physics Seminar Series, Cosmology Journal Club, Cosmology & Astrophysics Seminar at the University of Michigan, Kavli IPMU Astro Journal Club, and institute social coffee activities. |
| <b>Outreach</b>  | Panelist for r/askscience and YouTube AMA, Cosmology from Home 2023; scientist participant in Skype a Scientist, APS Physicists To-Go, and University of Michigan Science Communication Fellows.                                           |

## REFERENCES

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**Dr. Fabian Schmidt**

Ph.D. advisor

Max Planck Institute for Astrophysics

[fabians@MPA-Garching.MPG.DE](mailto:fabians@MPA-Garching.MPG.DE)

**Prof. Dragan Huterer**

Postdoctoral advisor

University of Michigan

[huterer@umich.edu](mailto:huterer@umich.edu)

**Prof. Elisabeth Krause**

External collaborator and mentor

University of Arizona

[krausee@arizona.edu](mailto:krausee@arizona.edu)